

Topic: API and Postman
Date: 24 January 2026

Recap:

HTTP module that is provided by Node.js
creating server and listening on a PORT

Express package
to create server similar to http module
under the hood, express utilizes http module for its working

Once endpoints are created they can be called for any response

Addition of middlewares
Middleware is a handler which can perform specific tasks before actually handling the request.

Middleware will be called before each of the request.

3 types of middlewares

1. In-built - they are provided by express itself
2. third party - installed from the npm registry and used for validations, logging and other functionalities
3. custom - built within the application

NOTE:

always call next() function on success else request will keep running forever

API

API stands for Application Programming Interface

it is a messenger which is helping two systems to interact with each other in a structured and defined way.

Defined here means:

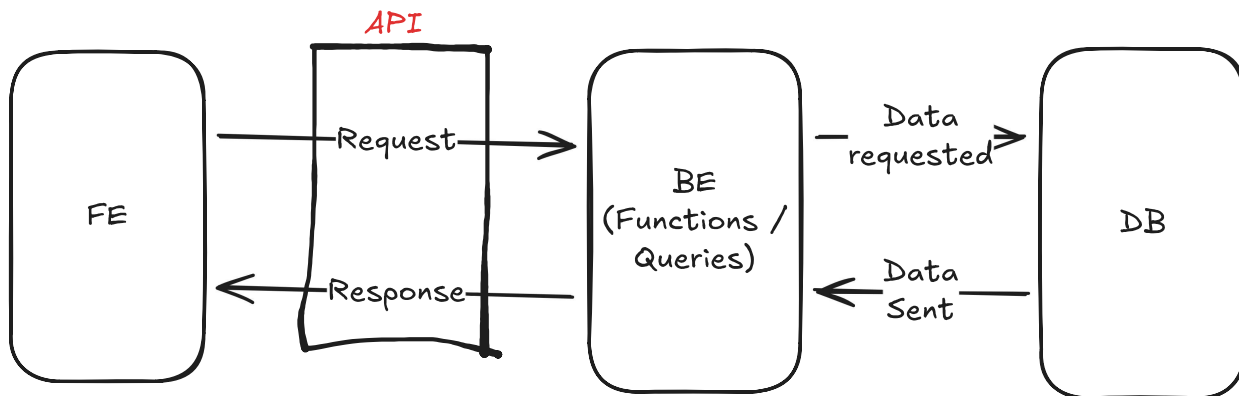
1. what you can ask for
2. how to ask for it
3. what you will receive as a response

Example of a Restaurant

1. Sitting on a table - Browser
2. Reading the menu - Exploring the frontend
3. Selection of meal - user interaction
4. Order the meal - API request sent from frontend
5. Order sent to kitchen - Request received at the backend
6. Preparation of Food - Gathering the data and processing it
7. Sending out the meal - Response sent to the frontend via API
8. Order received on the table - Frontend updates to the requested data

Idea is:

1. we never enter the kitchen - frontend never build logic
similar to db interaction or can access them
2. waiter is the only one with table to kitchen and kitchen to table access
- API is the only one who can take request from frontend to backend and response from backend to frontend



Frontend - Responsible to create the layout and consume the data from Backend

Backend - Process the request received from frontend and based on it gather data from the database if required

Database - it stored the data in table or document format

Real World Example

Login Form

1. User insert email and password
2. Clicks on Login
3. Frontend send the data to API
4. API sends the data to Backend
5. Backend checks the database
6. Database process and send a data to backend
7. Backend process the data received from database
8. Backend send the response to API
9. API send the response to frontend
10. Frontend process success / failure based on response

HTTP Methods

GET - reading the data , like getting products

POST - creating data, like adding products

PUT - replacing the data entirely, update product with new specifications

DELETE - removal of data, remove a product

PATCH - updating partial data, updating product with new name

Request Object

Information that a client send to the server which contains meta about how and what the request is?

1. req.method - HTTP method that is used
2. req.url - URL requested
3. req.params - URL parameters
4. req.query - Query string data
5. req.body - Data sent to the server from client
6. req.headers - extra info about the request

Response Object

sending a reply back to the client

1. res.send - using this we can send Text / HTML
2. res.json - using which we can send JSON data
3. res.status - to set the status code for the response
4. res.set - it is used to add response headers

Status Code

An identification number sent to the client to let them know what happened with the request.

2xx - Success - request was successful

3xx - Redirection - move request somewhere else

4xx - Client Errors / Unauthorized - person not available at the location at time of delivery

5xx - Server Error - Warehouse issues or delivery partner issues, or server bugs

200 -> Ok - data fetched / sent -> GET / PUT / PATCH

201 -> Created - New resource created -> POST

204 -> No Content -> DELETE

301 -> Moved Permanently

302 -> Temporary Redirect

304 -> Not Modified (Cache Response)

400 -> Bad Request

401 -> Unauthorized

403 -> Forbidden - No Permission

404 -> Not Found

409 -> Conflict

429 -> Too Many Request

500 -> Internal Server Error

502 -> Bad Gateway

503 -> Service Unavailable